

Machine Learning Major Capstone Project

Project Selection

Students must choose **one** of the project options below and complete it end-to-end, covering:

- Problem analysis
- Data collection & preprocessing
- Model development
- Implementation (from scratch where specified)
- Training & evaluation
- Testing & validation
- Deployment (if applicable)
- Final documentation and insights report

The selected project must demonstrate **strong conceptual understanding, technical implementation, and practical application.**

Option 1: Custom Regression Models Using NumPy (Sports Score Prediction)

Objective

Implement regression algorithms from scratch using NumPy to predict sports scores and understand underlying mathematical foundations.

Core Requirements

- Data preprocessing
- Implementation (NumPy only):
 - Linear Regression
 - Ridge (L2)
 - Lasso (L1)
 - Basic Decision Tree Regressor
- Performance evaluation (MSE, RMSE, R^2)
- Model comparison & analysis

Option 2: Custom Logistic Regression for Churn Prediction

Objective

Develop a logistic regression classifier from scratch using NumPy to predict customer churn in subscription-based platforms.

Core Requirements

- EDA & feature engineering
- Implementation of:
 - Sigmoid function
 - Log-loss (cross-entropy)
 - Gradient descent
- Class imbalance handling
- Evaluation using Accuracy, Precision, Recall, F1, ROC-AUC
- Benchmark with Scikit-learn

Expected Deliverables (For All Options)

- Complete Jupyter Notebook / Project Repository including:
 - Data preprocessing
 - Model implementation
 - Training results
 - Performance evaluation
- Final analytical report covering:
 - Model performance
 - Key insights
 - Business or practical implications
- Deployment demo (if applicable)